

Research article

Treatment of food waste contaminated by bioplastics using BSF larvae: Impact and fate of starch-based bioplastic films

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ABSTRACT

The use of Black Soldier Fly (BSF) larvae in the treatment of biowaste, including food waste, represents a promising new (waste) treatment option. In line with an increasing use of starch-based bioplastics in food packaging, (e.g. shopper films), food waste contamination by these polymers is expected to rise, but the fate of these materials and impact produced on the BSF treatment process remain to be clarified. In the present study, food waste contaminated by starch-based bioplastic film was treated using a BSF larvae process with the aim of investigating both the effect of bioplastics on process performance and the effect of BSF larvae on bioplastic degradation. Larvae treatment performance was assessed by monitoring substrate degradation process and larvae growth in terms of weight variation and development time. Bioplastic degradation (both in the larvae process and in a larvae-free control test) was assessed by means of visual inspection, Scanning Electron Microscopy (SEM), Fourier Transform InfraRed spectroscopy (FTIR), Differential Scanning Calorimetry (DSC) and ThermoGravimetric Analysis (TGA). The results obtained highlighted the absence of negative impacts of bioplastics on the BSF process, revealing a modestly higher degree of degradation in the larvae process compared to control test. The process however failed to achieve complete degradation of bioplastics, suggesting the need for additional post-processing treatments.

1. Introduction

In the framework of a Circular Economy and Bioeconomy, bioplastics are emerging on the global market as a more sustainable alternative to conventional fossil-based plastics as, by definition, they are made (fully or partially) from renewable resources (bio-based), are biodegradable or feature both properties. Indeed, based on these characteristics, bioplastics are capable of fostering enhanced circularity through use of renewable resources and biodegradation as an alternative end-of-life option, thus resulting in a reduction of widespread plastic pollution (Rosenboom et al., 2022).

Although bioplastics currently represent less than one percent of the global annual production of plastics, the market continues to expand, with a forecasted increase in bioplastic production from approx. 2.4 million tonnes in 2021 to 7.5 million tonnes in 2026 (European bioplastics, 2021).

Since 2015, the markets of Southern American and Asian countries (Brazil, India, China, and Japan) have featured the highest demand for bioplastics. In particular, following the banning of plastic bag disposal in

China, a dramatic increase in the production of PLA (Polylactic acid), PBAT (Polybutylene adipate terephthalate), and PBS (polybutylene succinate) is envisaged (Yamada and Fukumoto, 2021). Several other countries, including Japan, Malaysia, Singapore, and South Korea, plan to subsidize the production of bioplastics (Rosenboom et al., 2022). At European level, although a recent EU directive (EU, 2019/904) promoted a ban of single use plastics, including bio-based and biodegradable plastics, debate is still ongoing and a public consultation on the topic has been launched by the European Commission (European Commission, 2022).

As the market for bioplastics expands, the amount of bioplastic waste will also increase accordingly, thus implying the need for appropriate management of these materials.

With specific focus on bio-based and biodegradable plastics, the natural end-of-life option is represented by biodegradation. However, biodegradability of these polymers is strongly influenced by the environmental conditions under which degradation occurs, and compatibility with the biological process applied should be carefully assessed. For this purpose, recognized standards have been introduced by

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American Society for Testing and Materials (ASTM) and International Standards Organization (ISO) for use in assessing the biodegradability of materials under specific process conditions, such as composting and anaerobic digestion (Kale et al., 2007).

Thermoplastic starch (TPS)-based bioplastics are one of the most diffuse commercial bio-based and biodegradable bioplastics on the market today (RameshKumar et al., 2020). TPS-based bioplastics are produced by converting starch into TPS through a combined effect of temperature and shear, and subsequent blending with synthetic polymers to improve mechanical properties and resistance to moisture (Lopez et al., 2012).

Recently, the use of Black Soldier Fly (*Hermetia Illucens*) in the biological treatment of putrescible organic waste has proved to be a promising alternative to conventional biological treatments, with the advantage of converting organic waste into stabilised residues and larvae biomass suitable for recovery of material (e.g. animal waste, protein) and energetic (e.g. biodiesel) resources. BSF larvae have been used successfully in the treatment of a series of organic wastes, including manure (i.a. Liu et al., 2018; Zheng et al., 2019), agricultural waste (i.a. Lim et al., 2019; Palma et al., 2018), industrial waste (i.a. Norgren et al., 2020), food waste (Nguyen et al., 2015), sludge from wastewater treatment (Lalander et al., 2018), and digestate (Wee and Su, 2019). Moreover, promising results have been obtained from the treatment of high organic content wastewater (Grossule et al., 2020; Grossule and Lavagnolo, 2020; Popa and Green, 2012) introducing an effective feeding approach for liquid substrates (Grossule et al., 2021).

Large-scale facilities have been set up for the conversion of food waste into animal feed protein, oil, and plant fertilizers using BSF larvae. Founded in 2007, Enterra Feed Corporation (Canada) is thus currently able to recycle more than 130 tons of food waste per day (Waste and recycling, 2021). In South Africa, AgriProtein, founded in 2008, was the largest organic waste-based larvae farm, treating 250 tons daily to produce 4000 tons of protein meal, 3500 tons of oil, and 16,500 tons of fertiliser (Curtis, 2019). Following a recent expansion, the company set up a similar plant in California, with an intention to build 100 international fly farms by 2024 and 200 by 2027 (White, 2021). Other European examples includes Netherlands's Protix, France's Ynsect and Innova Feed.

In line with an increasing use of starch-based bioplastics in food packaging, such as shopper films, food waste contamination by these polymers is also expected to rise. Although the fate of bioplastics in conventional biological processes has been well studied (i.a. Balaguer et al., 2016; Du et al., 2008; Gómez and Michel, 2013; Ruggero et al., 2020), to our knowledge, the fate of these materials and impact produced on the BSF treatment process has not been evaluated to date.

The aim of the present study was to investigate both the effect of bioplastics on BSF larvae process performance and the effect of BSF larvae on bioplastic degradation. Food waste contaminated by starch-based bioplastic film was treated using the BSF larvae process. Larvae treatment performance was assessed by monitoring the substrate degradation process and larvae growth in terms of weight variation and development time. The results were compared with data present in literature to identify potential negative impacts of bioplastics on the larvae process.

To evaluate the effect of larvae process on bioplastic film degradation, a control test was performed, consisting of aerobic treatment, without larvae, of the food waste contaminated by bioplastics.

In both the larvae and control test, bioplastic degradation was evaluated at the beginning of and throughout the test by means of visual inspection, Scanning Electron Microscopy (SEM), Fourier-transform infrared spectroscopy (FT-IR), Differential Scanning Calorimetry (DSC) and Thermogravimetric Analysis (TGA).

2. Materials and methods

2.1. Research program

BSF larvae were used to treat food waste contaminated with starch-based bioplastic films (L test). A control test without larvae was performed, consisting in treatment of the bioplastic contaminated food waste under aerobic conditions (C test) (Fig. 1). Both tests were carried out in triplicate.

2.2. Substrate and waste matrix preparation

The waste matrix in each test consisted in a mixture of 7 kg food waste (mix of vegetables, pasta, meat, etc), 0.2 kg wood branches, and 0.049 kg starch-based bioplastic film. Food waste served as the only feeding substrate for larvae, while wood branches (about 5 cm length, 0.2–0.5 cm diameter), difficult to metabolise by larvae (Lohri et al., 2017), were added as structured material to increase porosity of the matrix and facilitate air circulation. Bioplastic concentration in the waste matrix was approx. 0.7% (in weight of wet basis) of food waste, in line with the typical bioplastic contamination of food waste reported by Italian composting plants (Consorzio Italiano Compostatori, 2017) and with previous studies on compostability of bioplastics (Lavagnolo et al., 2020). Bioplastic film was obtained by cutting Mater-Bi® supermarket bags into pieces of approx. 5 × 5 cm. Mater-Bi® biopolymer, labelled as compostable, is a thermoplastic comprised of 20% starch and 70% PBAT (Poly-Butylene-Adipate-co-Terephthalate), blended with 10% additives (Elfehri Borchani et al., 2015).

2.3. Equipment and operational conditions

Tests were performed in batch reactors, consisting in 20 L glass vessels. Each reactor was covered with a tight mesh net to avoid oviposition of other flies, wrapped with insulating material and equipped with a temperature probe and an aeration system, providing a flowrate to the matrix of 60 L/hour (Fig. 2). Aeration rate was selected based on optimum aeration conditions for BSF larvae, as suggested by Palma et al. (2018). The experiment was carried out in a thermally insulated room under the same environmental conditions suggested by Grossule and Lavagnolo (2020): temperature range 25–30 °C; photoperiod Light/Dark of 18/6 h.

2.4. Feeding, operation and monitoring

All reactors were filled with the waste matrix at the beginning of the test. For the larvae test (L), the matrix was inoculated with approximately 7650 larvae, approx. 5-Day old. The quantity of larvae inoculated was selected based on the following assumptions: 7 kg of food waste was the only feeding substrate, optimum food waste feeding rate is approx. 61 mg/d/larva, expected larvae development time is approx. 15 days (Diener et al., 2009).

Daily monitoring was carried out by manually turning the waste matrix and recording waste matrix temperature. Food waste, larvae and

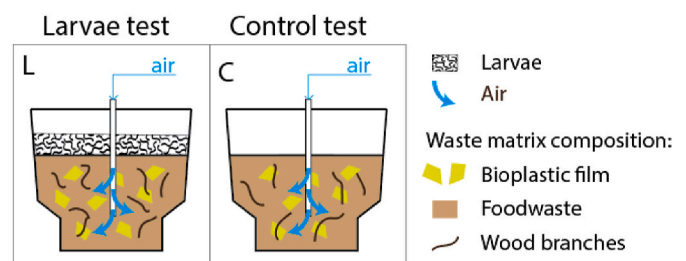


Fig. 1. Graphical description of the research scheme.

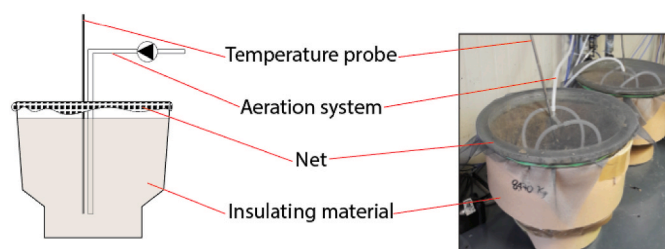


Fig. 2. Testing reactors arrangement.

bioplastic film were sampled every five days until 50% of larvae had progressed to the prepupae stage, and again at the end of the test when all larvae had reached the prepupal stage. Prepupae were easily recognized by skin darkening.

Food waste and food waste residues at the end of the process were weighed and analysed for Total Solids (TS), pH, ammonia nitrogen (N-NH_4^+), and Respirometric Index after 4 days (RI_4).

Larvae were washed and individually weighed before returning to the reactor. Larvae process performance was evaluated by monitoring larvae growth and calculating Food conversion rate (FCR), Bioconversion ratio (BCR) (both on dry basis; Lalander et al., 2019; Rehman et al., 2017), and efficiency of conversion of digested food (ECR, on wet basis) (Meneguz et al., 2018), using the following equations:

$$\text{FCR} \left(\frac{g_{\text{substrate}}}{g_{\text{larvae}}} \right) = \left(\frac{\left(\frac{\text{TS}(\%)}{100} \right) * (g_{\text{substrate initial}}) - \left(\frac{\text{TS}(\%)}{100} \right) * (g_{\text{substrate final}})}{\left(\frac{\text{TS}(\%)}{100} \right) * (g_{\text{biomass larvae final}})} \right) \quad 1$$

$$\text{BCR} (\%) = \left(\frac{\left(\frac{\text{TS}(\%)}{100} \right) * (g_{\text{biomass larvae final}})}{\left(\frac{\text{TS}(\%)}{100} \right) * (g_{\text{substrate initial}})} * 100 \right) \quad 2$$

$$\text{ECR} \left(\frac{g_{\text{larva}}}{g_{\text{substrate}}} \right) = \left(\frac{(g_{\text{biomass larvae final}})}{(g_{\text{substrate initial}}) - (g_{\text{substrate final}})} \right) \quad 3$$

The results obtained were compared with literature data to assess the potential negative impacts of bioplastic film on larvae process. Bioplastic samples were washed (with distilled water), dried (at room temperature), and analysed by means of visual inspection, SEM, FT-IR, DSC, and TGA.

2.5. Analytical procedure and interpretation

Food waste was analysed according to Standard Italian Methods (IRSA-CNR). TS was measured by gravimetric determination after 24 h drying at 105 °C. pH was measured in the eluate of a batch test, prepared according to UNI-EN standards, conducted at Liquid/Solid ratio of 10. Ammonia nitrogen was measured using a distillation-titration procedure. RI_4 was performed using Sapromat E.

Images of bioplastic film were collected using SEM, FEI ESEM Quanta 200 apparatus, after being metalized with a 10 nm layer of gold at a spatial resolution of 300 μm , 100 μm , 50 μm and 10 μm .

In the FT-IR analysis procedure, samples are subjected to infrared radiation (IR) and the FT-IR spectra is obtained by plotting the absorbance and/or transmittance of energy by chemical bonds of the material against the wavenumber. Peak wavenumbers in the FT-IR spectra provide information relating to chemical composition of the material (Nandiyanto et al., 2019). Variation of peak intensity elicited by bioplastics at different treatment times provided qualitative information indicating chemical changes to the polymeric structure during the degradation process.

FT-IR spectra were collected in attenuated total reflection mode (ATR) using a Thermo Scientific™ Nicolet™ iS™50 FT-IR Spectrometer

with 2 cm^{-1} spectral resolution.

DSC analysis allows physical changes of the material in response to temperature to be studied by measuring heat flow absorbed or released by the material against temperature variation. The material undergoes a first heating cycle (to erase the thermal history the material is subjected to during processing), cooling, and a second heating cycle. Endothermic (heat absorbed) and exothermic (heat released) heat flows may be detected during the heating and cooling cycle. If the material is semi-crystalline, the crystalline fraction undergoes melting (at melting temperature, T_m) during the heating cycles (endothermic reaction), and crystallization (at crystallization temperature, T_c) during cooling cycle (exothermic reaction). The amorphous fraction, on the other hand, undergoes glass transition, passing from a rigid glassy state at temperatures below T_g to rubbery solid state at temperatures above T_g . Differential Scanning Calorimetry (DSC) was performed using a heat-cool-heat cycle, from 40 °C to 200 °C, with a heating rate of 20 °C/min and cooling rate of 10 °C/min.

TGA analysis yielded thermal characterisation of the material by measuring change in mass as a function of temperature. Slight variations in weight observed at low temperatures are usually due to the presence of moisture or volatile solvents, while strong variations are caused by decomposition of the organic fraction of the polymer itself. Decomposition temperatures of the organic fraction of the sample and the inorganic residual are recognizable in a thermogram. Analysis was carried out in an oxidizing atmosphere (in air) to obtain thermo-oxidative decomposition, from 40 °C to 900 °C with a heating rate of 20 °C/min.

3. Results and discussion

3.1. Degradation process

Mean temperature, moisture, and pH are amongst the most significant factors implicated in control of biodegradation of bioplastics. Hydrolysis reaction and microbial growth rates accelerate in line with temperature increase (until thermophilic conditions are reached) and under moisture-rich conditions (Kale et al., 2007). In starch-based polymers, starch is hydrolysed under alkaline conditions (Sakkara et al., 2020).

Average temperature, moisture content, pH, and ammonia content variation in both L and C tests are illustrated in Fig. 3A. Starting from the 5th day, temperature increased above ambient temperature in both L and C tests, achieving mesophilic conditions. L test displayed the highest temperature, up to 45 °C, due to heat release by larval metabolism and the generation of movement-related friction (Bonelli et al., 2019). No significant variations were observed in moisture content and pH of either L or C tests until the 15th day, ranging between 47 and 52% and between 3.5 and 4, respectively. Lower moisture content and significantly higher pH values were detected at the end of both tests. In particular, alkaline conditions were achieved in L test, generally as the result of larval release of ammonia-rich metabolic residues (Ma et al., 2018). Indeed, ammonia concentration variation (Fig. 3A) increased significantly in L test during larval growth phase, up to 6800mgN/kgTS on 15th day, in line with previous studies (Liu et al., 2020). Ammonia decreased during prepupation up to the end of the test due to reduced production of metabolic residues and volatilisation of ammonia gas at high pH conditions. Conversely, ammonia concentration decreased in C test from day 5. Larvae process conditions were seen to be more favourable, in terms of higher moisture, temperature, and pH for bioplastic degradation compared to those displayed by C test.

Food waste consumption and stabilisation were evaluated in terms of TS mass and RI_4 reduction (Fig. 3B). TS mass balance was computed subdividing the initial load into substrate residue, prepupal biomass and gas emissions (Fig. 3B), where gas emissions were calculated as the difference between initial and final TS mass. Larvae treatment displayed higher TS reduction compared to C test. 1.67 kgTS of food waste was reduced to 0.38 kgTS in L test and to 0.76 kgTS in C test, achieving a

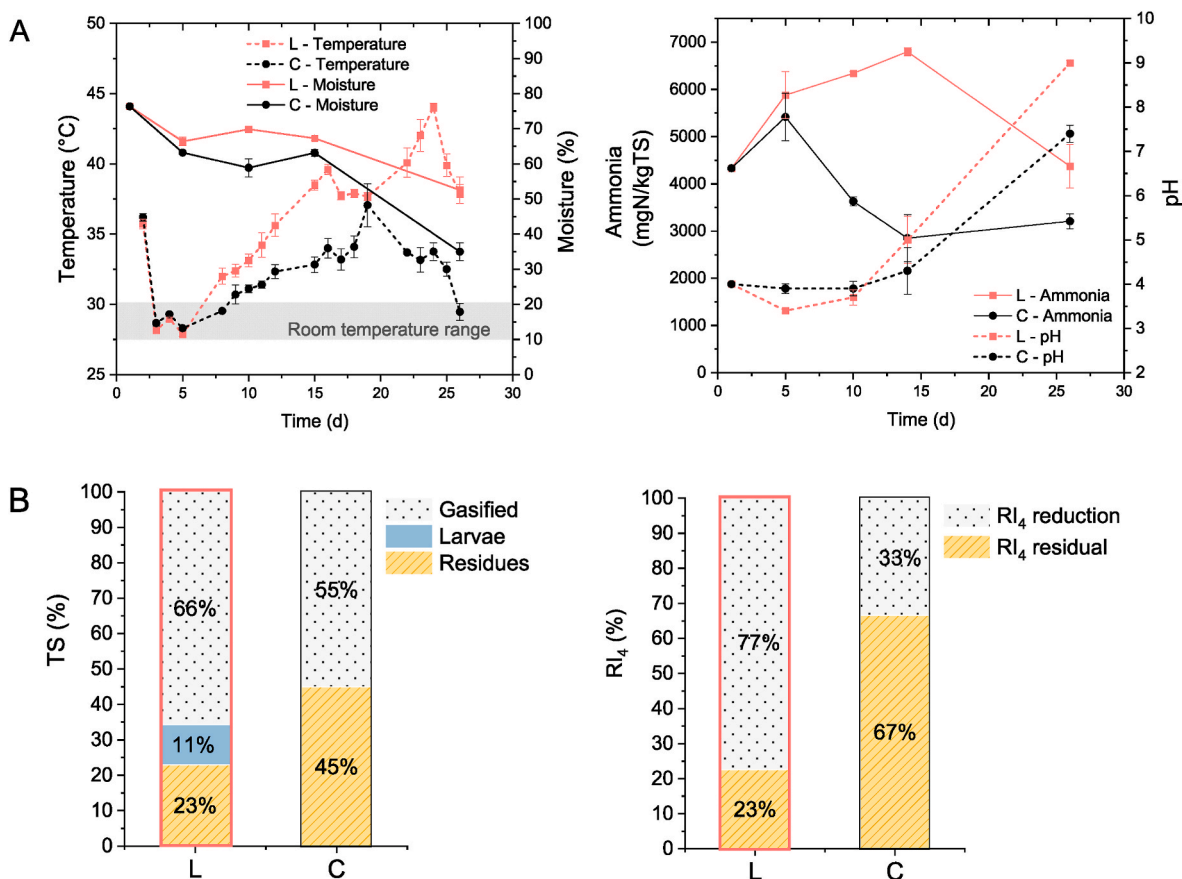


Fig. 3. Food waste analysis in both L and C tests: average variation of temperature, moisture content, pH, ammonia content (A); TS mass balance and RI₄ reduction (B). (L = larvae test; C = control test).

reduction rate of approx. 77% and 55%, respectively. A higher waste reduction was obtained in L test compared to previous studies, in which TS reduction ranged between 46.7 and 61.9% (Lalander et al., 2019). In L test, $11 \pm 1.87\%$ of the initial TS was converted into larvae dry biomass, in line with previous studies (Diener et al., 2009; Grossule et al., 2020).

Initial RI₄ value measured on food waste was approx. $105 \pm 2 \text{ mgO}_2/\text{gTS}$. L test displayed higher RI₄ reduction (77%) compared to C test (33%) at the end of the experiment, with final RI₄ values of 24 ± 1 and $70 \pm 2 \text{ mgO}_2/\text{gTS}$ in L and C test, respectively. Similar RI₄ reduction was obtained by Grossule et al. (2020) (approx. 82%) by treating wheat bran

and brewers' spent grain with larvae. Compared with data present in literature, the results obtained for food waste consumption and stabilisation demonstrated the absence of a negative impact of bioplastic film on larvae process performance.

3.2. Larvae growth

Average larval wet weight variation measured during L test is reported in Fig. 4A. To evaluate the impact of bioplastic contamination on larval growth, maximum wet weight, and development time (when at least 50% larvae prepupae had emerged), were compared with literature

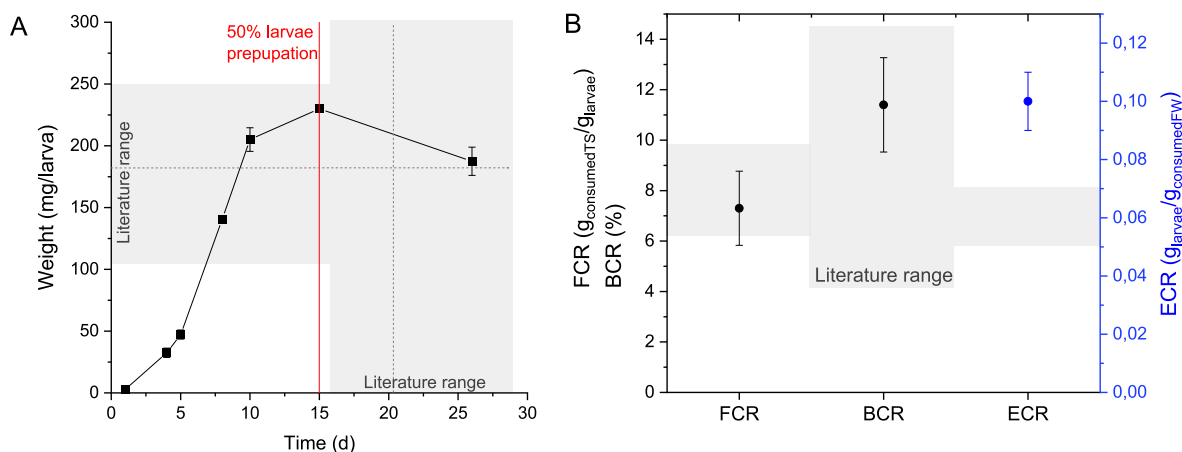


Fig. 4. Average larval wet weight variation measured during the experiment (A) and obtained Food conversion rate (FCR), Bioconversion ratio (BCR) and efficiency of conversion of digested food (ECR) in L test (B). Literature ranges and mean values are indicated by grey area and dotted lines, respectively.

data, obtained by feeding larvae with food waste or similar (Table 1). Literature ranges for maximum wet weight and development time are also indicated in Fig. 4A.

Larvae displayed an exponential growth, achieving maximum larval wet weight of approx. 187 mg and 50% prepupation over a period of 15 days. These findings are in line with literature data and amongst the highest wet weight values and shortest development times recorded in similar laboratory experiments conducted previously (Fig. 4A, Table 1).

On the basis of food waste reduction and larvae TS content of 35%, FCR, BCR, and ECR were calculated according to Eqs. (1)–(3) and compared with literature data for similar larvae feeding substrate (Fig. 4B). The mean FCR of approx. 7.3 ± 1.47 $g_{\text{substrate}}/g_{\text{larva}}$ is comparable with values obtained by Somroo et al. (2019) and Rehman et al. (2017) corresponding to 6.2 and 9.8 $g_{\text{substrate}}/g_{\text{larva}}$, respectively. The obtained mean BCR of $11.4 \pm 1.87\%$ was within the range obtained by Lalander et al. (2019) and characterized by extremes of 4.1% and 14.2% (depending on the content of abattoir waste). ECR was approx. 0.1 ± 0.01 $g_{\text{larva}}/g_{\text{substrate}}$, therefore higher than values reported by Meneguz et al. (2018) using fruit and vegetable substrates: 0.07 $g_{\text{larva}}/g_{\text{substrate}}$ and 0.05 $g_{\text{larva}}/g_{\text{substrate}}$, depending on vegetable content (Fig. 4B).

The findings obtained confirmed that the presence of bioplastic film in food waste produced no significant negative impact on larval growth and substrate consumption.

3.3. Bioplastic degradation

The process of degradation of bioplastic film was initially investigated by means of visual inspection (Fig. 5A). No significant change in size was present in either L or C test. Conversely, a significant difference in colour was observed. In particular, from the 10th day, bioplastics in L test changed from an initial transparent light green colour to an opaque, intense warm brown. This could be explained by more favourable conditions for bioplastic degradation in L test from the 10th day, as previously discussed (Fig. 3).

SEM micrographs of bioplastic film are reported in Fig. 5B. Material appearance was initially visible as a heterogenous microstructure characterized by circular spots, typically attributed to starch (Aldas et al., 2020; Ruggero et al., 2020), dispersed in a polymeric matrix assumed to be mainly PBAT (Muthuraj et al., 2015). Fig. 5B illustrates the presence of a surface smoothing process in C test, with gradual disappearance of the granular shape, revealing an increasingly visible porous base. An identical result was reported by Ruggero et al. (2020) and Zhai et al. (2020), and justified by the progressive degradation of circular starch spots. A faster smoothing process of bioplastic surface was already visible in L test on the 5th day. From the 10th day, jagged engraving on the material was visible, likely caused by friction between

the larvae body and bioplastic film.

FT-IR analysis provided information relating to the chemical change in bioplastic film during the degradation process. Fig. 6 illustrates the most significant spectra obtained at the beginning and the end of both L and C tests. Only wavenumbers in the range $700\text{--}3000\text{ cm}^{-1}$, featuring presence of the main diagnostic peaks, are illustrated. Peaks typically attributed to starch and PBAT chemical bonds are reported in the same figure. The main characteristic peaks of starch are those occurring between 1017 and 1158 cm^{-1} (glycosidic bond) (Brandelero et al., 2011; Zhai et al., 2020), and at 2922 cm^{-1} due to C–H stretching (Abdullah et al., 2018).

Typical PBAT peaks are attributed to methylene group at 727 cm^{-1} , C–O (bond) at 1269 cm^{-1} , C–H at 1458 cm^{-1} , C–O (carbonyl ester group) at 1711 cm^{-1} (Brandelero et al., 2011; Elfehri Borchani et al., 2015; Pagno et al., 2020; Ruggero et al., 2020; Zhai et al., 2020).

At 1017 cm^{-1} a peak due to C–O could be attributed to both starch and PBAT (Brandelero et al., 2011; Elfehri Borchani et al., 2015; Zhai et al., 2020).

No remarkable changes were observed in the chemical structure of bioplastic. However, slight differences were detected between spectra, in correspondence with starch attributed peaks: reduction of the peaks between 1017 cm^{-1} and 1158 cm^{-1} suggested cleavage of the glycosidic bonds of starch (Azevedo et al., 2003).

Findings were confirmed by DSC and TGA analysis (Fig. 7A–B).

Using Differential Scanning Calorimetry analysis, melting peaks were detected in bioplastic film both at the beginning and end of L and C tests between 120 and $126\text{ }^{\circ}\text{C}$ (T_m). The results obtained were in line with values present in literature for TPS containing PBAT, with typical T_m between 110 and $126\text{ }^{\circ}\text{C}$ (Aldas et al., 2020; Elfehri Borchani et al., 2015).

The melting temperature of the polymer increased at the end of both L and C tests, with T_m values of $126.13\text{ }^{\circ}\text{C}$ and $123.35\text{ }^{\circ}\text{C}$, respectively (Fig. 5A). Higher T_m values suggested an increased stability of the crystalline phase of the polymer due to degradation of the amorphous regions of the molecular structure, consistent with previous studies (Fu et al., 2020; Kijchavengkul et al., 2010). In the L test, the highest increase in T_m pointed to a slightly higher degradation of the amorphous region compared to C test.

In TGA thermogram (Fig. 5B), weight variations at temperatures of approx. $300\text{ }^{\circ}\text{C}$ and just above $400\text{ }^{\circ}\text{C}$ were attributed to starch and PBAT, respectively, in line with the findings of previous studies (Al-Itry et al., 2012; Elfehri Borchani et al., 2015; Pagno et al., 2020; Ruggero et al., 2020). Weight (%) equal to zero at temperatures above $550\text{ }^{\circ}\text{C}$ suggests the absence of an inorganic fraction in the sample. Slight changes were detected when comparing the results yielded by TGA. At approx. $300\text{ }^{\circ}\text{C}$, weight variation due to the presence of starch was lowest in L test, indicating a higher contribution of BSF larvae to starch degradation compared to C test. Results were consistent with the findings of a study conducted by Ruggero et al., 2020: PBAT was the most resistant component of the starch-based blend, the biodegradation of which was strictly dependent on both high temperature and moisture values.

The evaluation of bioplastic degradation by means of visual inspection, SEM, FT-IR, DSC and TGA revealed that BSF larvae process provided more favourable conditions for bioplastic degradation, which was improved, but without achieving remarkable results. It can be mainly attributed to the relatively low temperature at which BSF process occurred: maximum temperature was in the range $40\text{--}45\text{ }^{\circ}\text{C}$ and maintained only for three days. Conversely, the compostability of the tested bioplastic, thus its degradation under composting conditions, is ensured by the thermophilic phase of almost three weeks, at the beginning of the composting process, with temperature ranging between 55 and $60\text{ }^{\circ}\text{C}$.

Further studies should be carried out to assess the bioplastic degradation performance when constant process temperature is maintained at $40\text{ }^{\circ}\text{C}$, bioplastic is recirculated in the BSF process at an extended treatment time, or when combining BSF treatment with final refining of

Table 1

Larvae maximum wet weight and development time (when at least 50% larvae prepupae had emerged) obtained in previous studies by feeding larvae with food waste or similar and obtained in the present study.

Substrate type	Max wet weight (g)	Development Time (days)	Source
Food waste	0.212	19	Lalander et al. (2019)
Abattoir waste	0.248	17	
Abattoir waste + fruit and vegetables	0.252	17	
Fruit and vegetables	0.104	20.2	Meneguz et al. (2018)
Fruit	0.109	22	
Kitchen waste	0.173	23.8	Nguyen et al. (2013)
Fruit and vegetables	0.123	28.6	
Food waste	0.227		Nguyen et al. (2015)
Fruit and vegetables	0.170		
	0.179	21	Mean
	0.104	17	Min
	0.252	28.6	Max
Food waste	187 ± 11	15	This study

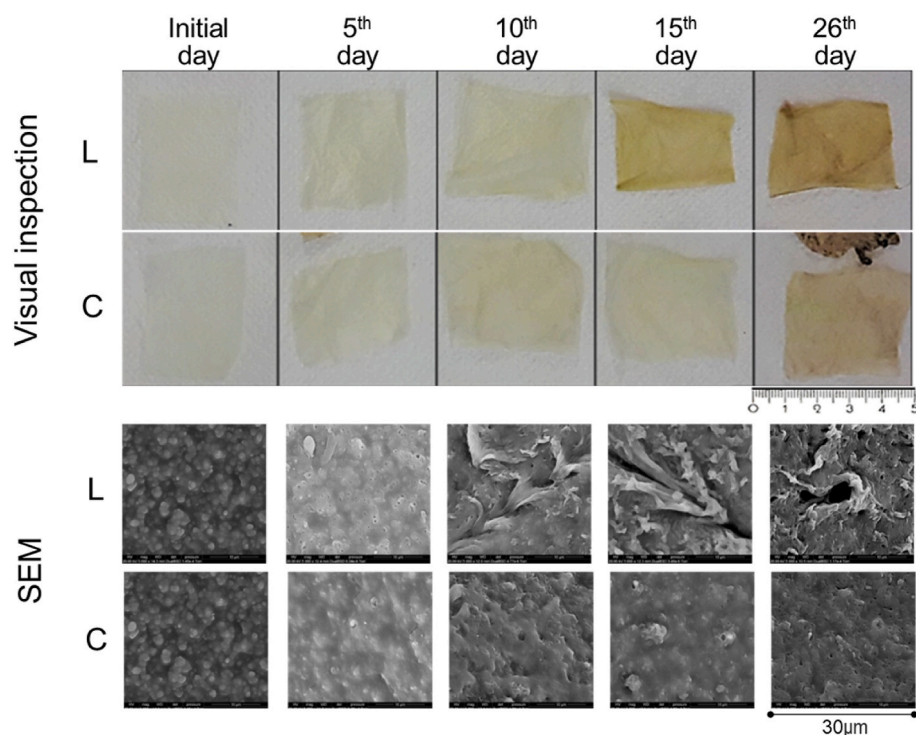


Fig. 5. Visual and magnified appearance of bioplastic film obtained by means of visual inspection and SEM analysis (x5000) at the beginning and during the process in both L and C tests.

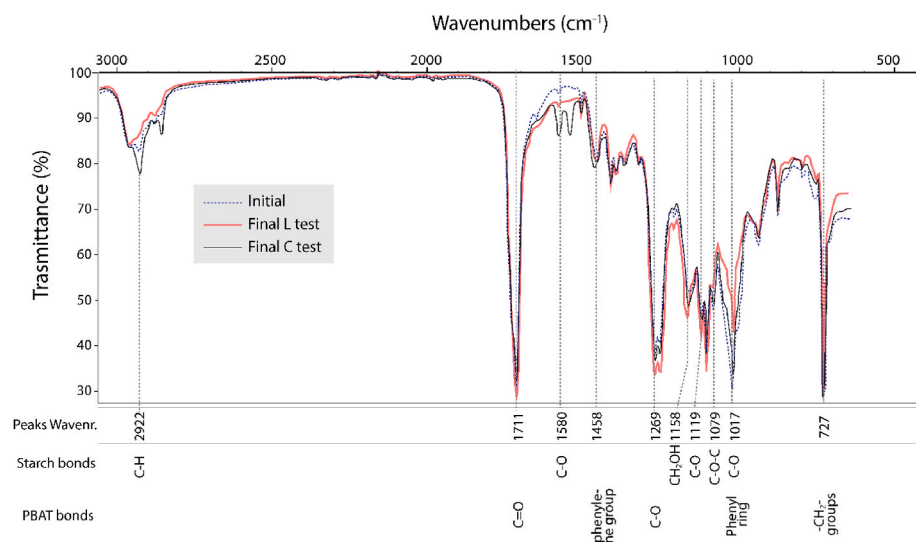


Fig. 6. Comparison of FT-IR spectra obtained from starch-based biopolymer film at the beginning and end of treatment in both L and C tests. (L = larvae test; C = control test).

residues by means of composting or anaerobic digestion. Further studies should also assess the potential formation of microplastics during treatment and accumulation in the larval biomass.

4. Conclusions

Food waste contaminated by starch-based bioplastic film was treated using a BSF larvae process with the aim of investigating both the effect of bioplastics on process performance and the effect of BSF larvae on bioplastic degradation. Based on the results obtained, the following conclusions may be drawn.

- In general, BSF process performance was not negatively affected by the presence of bioplastic film: larvae tests displayed some of the highest larval wet weight values and shortest development times recorded in similar laboratory experiments conducted previously (187 mg and 50% prepupation within 15 days). Food waste reduction (77% TS removal) and stabilisation, as well as the performance of substrate conversion into larval biomass (FCR, BCR, ECR values) were in line with literature data.
- Larvae process conditions were found to be more favourable in terms of higher moisture, temperature, and pH to bioplastic degradation compared to those displayed by C test.

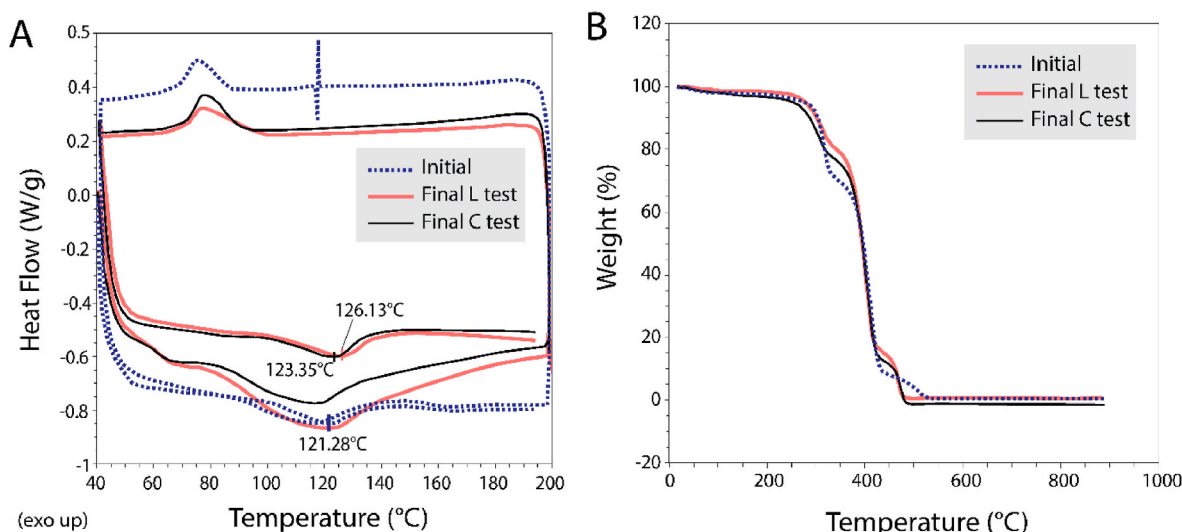


Fig. 7. DSC curve (A) and TGA thermogram (B) obtained from Mater-Bi bioplastic film at the beginning and end of treatment in both L and C tests. (L = larvae test; C = control test).

c) Bioplastic degradation was evaluated at the beginning and throughout the test. According to the results obtained by means of visual inspection, SEM, FT-IR, DSC and TGA analysis, a modest improvement in bioplastic degradation occurred under BSF larvae treatment compared to control test. However, that complete degradation of the biofilm was not achieved after a 26-day larval life cycle.

Credit author statement

Valentina Grossule: Conceptualization, Methodology, Laboratory activities, Data elaboration, Data interpretation, Writing Original draft. **Silvia Zanatta:** Laboratory activities, data interpretation, paper review. **Michele Modesti:** Data interpretation, paper review. **Maria Cristina Lavagnolo:** Data interpretation, paper review.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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